Control Variates for Bayesian Estimation of a GARCH Model

TD project

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Abstract

This note implements a random-walk Metropolis sampler for a Bayesian GARCH(1,1) model and compares plain MCMC estimates with zero-variance control variates. The first controls are the linear-polynomial controls of Mira, Solgi and Imparato (2013). Larger polynomial dictionaries are then handled either by ordinary least squares or by a Lasso screening step inspired by Leluc, Portier and Segers (2021). Numerical error is assessed by independent repeated runs and by batch-means standard errors.

1 Model and Data

For returns r_t, the model is

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \qquad h_t = \omega_1 + \omega_2 r_{t-1}^2 + \omega_3 h_{t-1}.$$

The parameter vector is $\omega = (\omega_1, \omega_2, \omega_3)$. We use independent centered normal priors on ω_i, truncated to

$$\omega_1 > 0, \qquad \omega_2 > 0, \qquad \omega_3 > 0, \qquad \omega_2 + \omega_3 < 0.995.$$

The likelihood is the usual Gaussian quasi-likelihood, up to constants,

$$\ell(\omega) = -\frac{1}{2} \sum_t \left\{ \log h_t + \frac{r_t^2}{h_t} \right\}.$$

Two datasets are used. The first is simulated from

$$\omega^\star = (0.04, 0.08, 0.88)$$

with 600 observations. The second is daily EUR/USD data from 2023-01-01 to 2024-12-31, converted to percentage log-returns. The exchange-rate data were downloaded from Frankfurter and saved locally in the project folder.

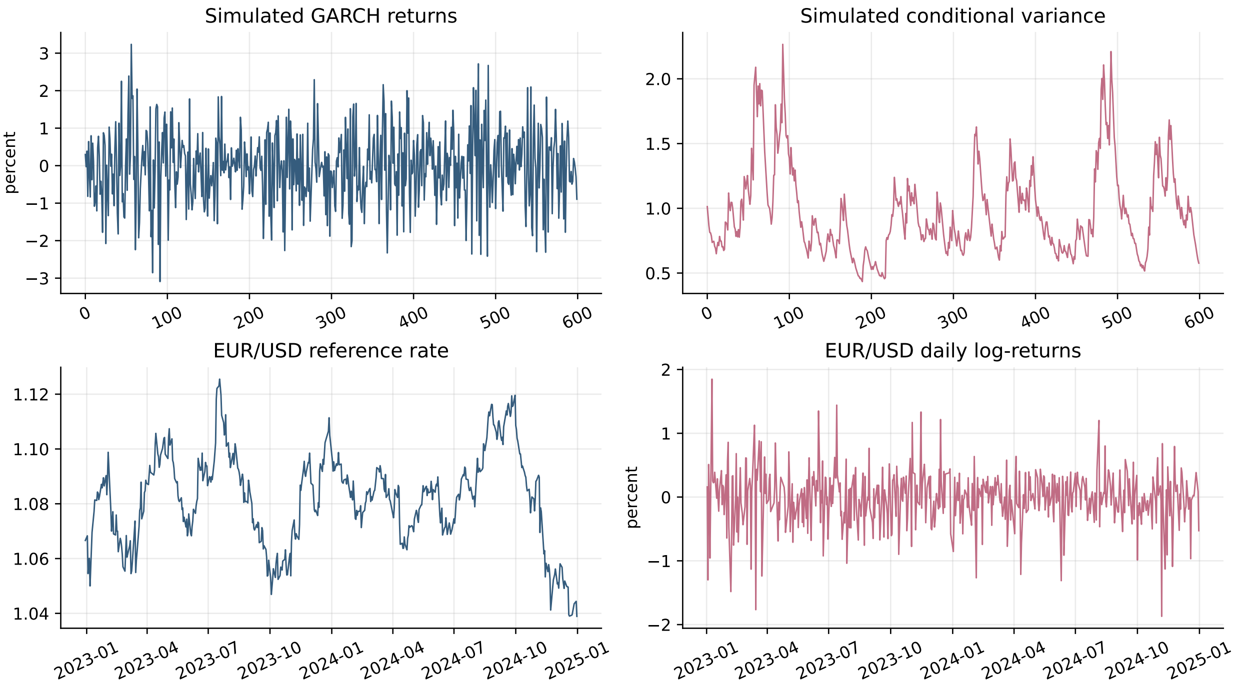

Figure 1: Simulated GARCH series and EUR/USD exchange-rate data.

2 Random-Walk Metropolis

The sampler is a random-walk Metropolis algorithm. To avoid negative proposals and to make the target support unconstrained, the chain is run on $\theta \in \mathbb{R}^3$, mapped to ω by

$$\omega_1 = e^{\theta_1}, \qquad \omega_2 = \rho \frac{e^{\theta_2}}{1 + e^{\theta_2} + e^{\theta_3}}, \qquad \omega_3 = \rho \frac{e^{\theta_3}}{1 + e^{\theta_2} + e^{\theta_3}}, \quad \rho = 0.995.$$

The transformed target includes the Jacobian of this map. The proposal is

$$\theta' = \theta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \operatorname{diag}(s_1^2, s_2^2, s_3^2)).$$

The diagonal scale was tuned by a short pilot chain. Each numerical experiment used 10 independent chains, 4000 Metropolis iterations per chain, burn-in 800, and thinning by 2.

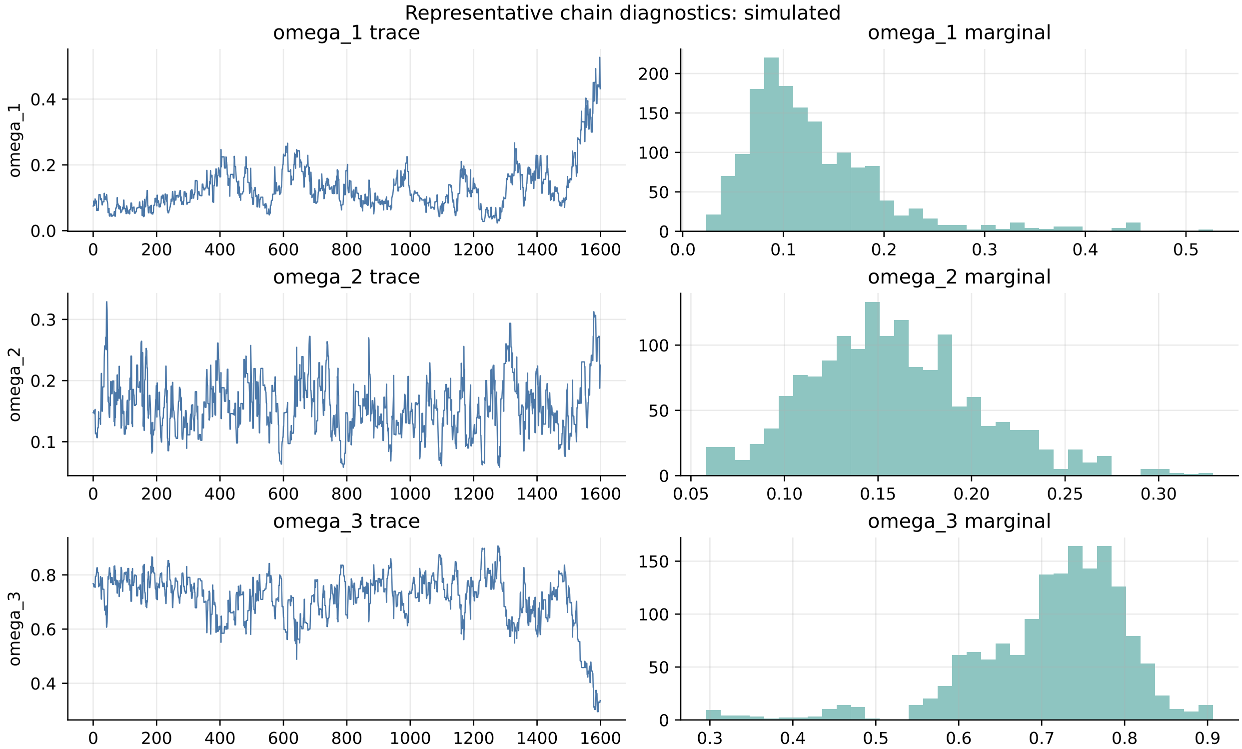

Figure 2: Representative posterior chain for the simulated dataset.

3 Zero-Variance Control Variates

For a target density $\pi(x)$, Mira, Solgi and Imparato define

$$z(x) = -\frac{1}{2}\nabla \log \pi(x).$$

For a polynomial trial function P, the controlled integrand is

$$\widetilde{f}(x) = f(x) - \frac{1}{2}\Delta P(x) + \nabla P(x)^\top z(x).$$

If $P(x) = a^\top x$, then $\Delta P = 0$ and

$$\widetilde{f}(x) = f(x) + a^\top z(x).$$

The optimal a is obtained by minimizing the empirical variance. Equivalently, regress $f(x_i)$ on the control vector $z(x_i)$; if the regression coefficient is $\hat{\beta}$, the controlled estimate is

$$\widehat{\mu}_{\text{cv}} = \bar{f} - \bar{z}^\top \hat{\beta}.$$

In this project $x = \theta$, while $f_j(\theta) = \omega_j(\theta)$, because the posterior summaries are desired in the original GARCH parameters.

4 Larger Dictionaries and Lasso

For all monomials up to total degree q in dimension $d = 3$, the number of non-constant polynomial controls is

$$\binom{d+q}{d} - 1.$$

This grows quickly: degree 1 gives 3 controls, degree 2 gives 9, and degree 5 gives 55. Ordinary least squares on all controls can become unstable because the design matrix becomes highly collinear; it also becomes expensive when d or q is larger.

Following Leluc, Portier and Segers, we use Lasso as a screening step. The implementation standardizes the controls, computes a Lasso path by coordinate descent, selects the penalty by BIC, and finally refits ordinary least squares only on the selected controls. This is the “ZV-Lasso-5” method in the tables.

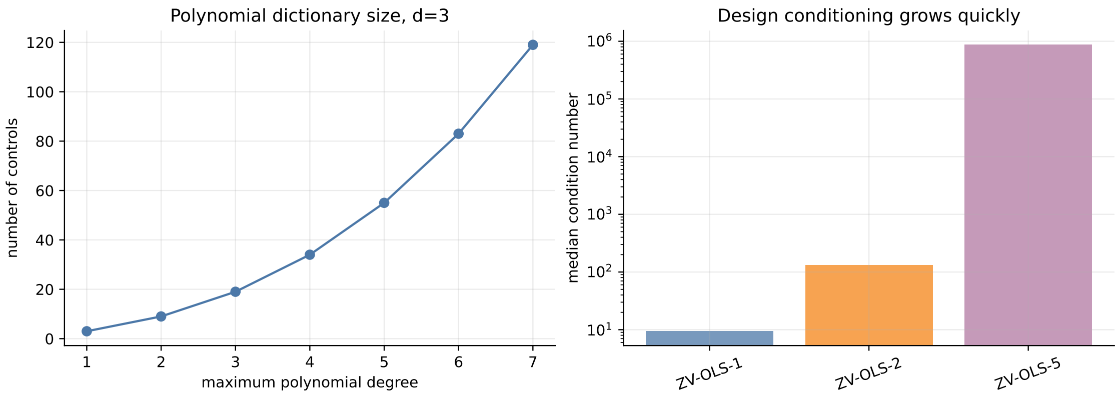

Figure 3: Number of polynomial controls and conditioning of the OLS design.

5 Numerical Results

The boxplots show the repeated-run variability of each posterior mean estimate. For the simulated case, the dashed line is the data-generating value; finite-sample posterior means need not equal it. For EUR/USD, the dashed line is the pooled raw-MC mean, used only as a visual reference.

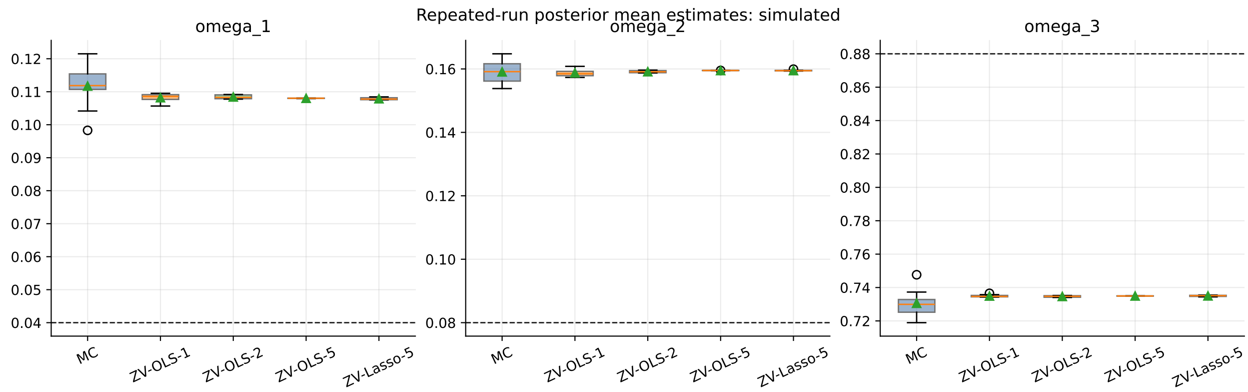

Figure 4: Repeated-run estimates for simulated GARCH data.

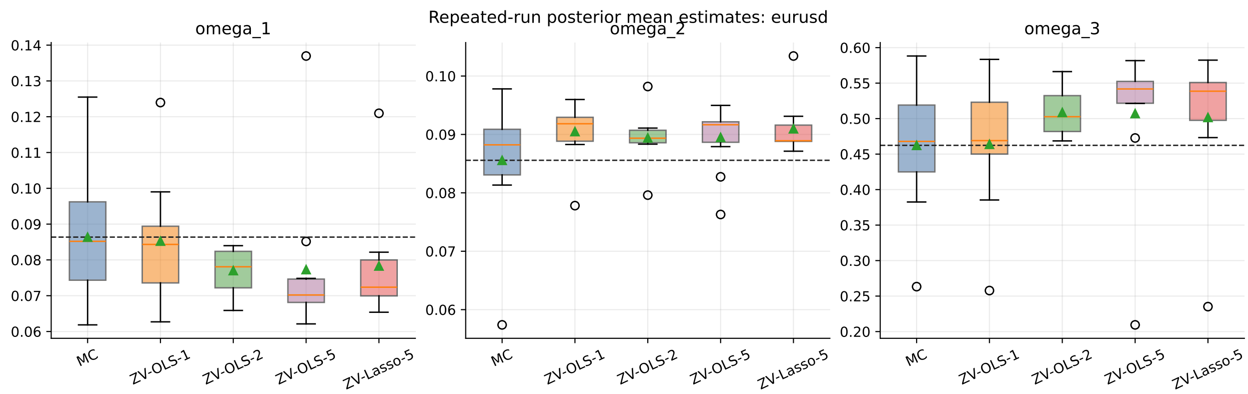

Figure 5: Repeated-run estimates for EUR/USD log-returns.

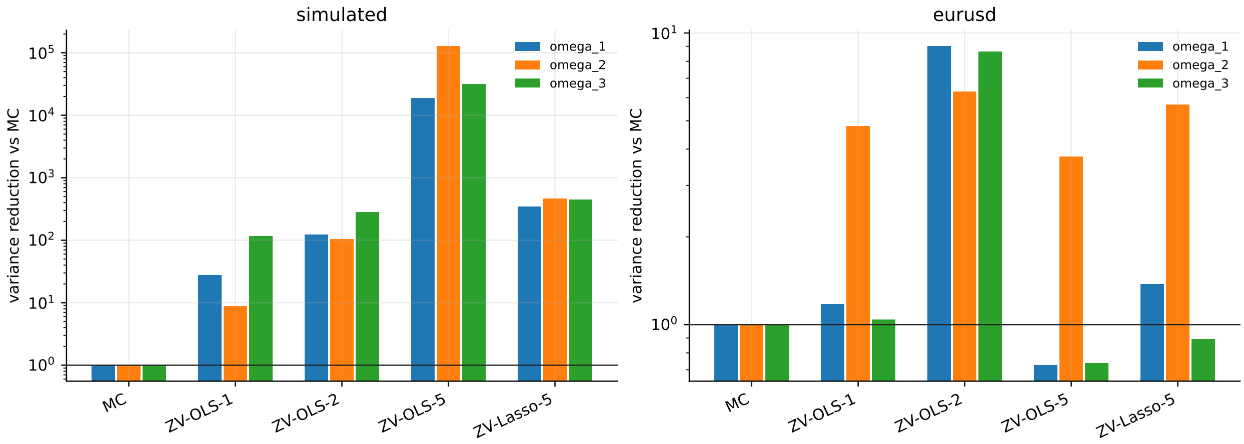

Figure 6: Empirical variance reduction relative to raw Monte Carlo. Values above 1 improve over MC.

5.1 Simulated Data

data	param.	method	mean	sd across runs	rmse/ref.	var. red.	avg. selected
simulated	omega_1	MC	0.1117	0.0066	0.0720	1.0000	0.0000
simulated	omega_1	ZV-Lasso-5	0.1079	3.58e-04	0.0679	344.4537	25.1000
simulated	omega_1	ZV-OLS-1	0.1082	0.0013	0.0682	27.7349	3.0000
simulated	omega_1	ZV-OLS-2	0.1084	5.97e-04	0.0684	123.9140	9.0000
simulated	omega_1	ZV-OLS-5	0.1080	4.83e-05	0.0680	1.89e+04	55.0000
simulated	omega_2	MC	0.1591	0.0037	0.0792	1.0000	0.0000
simulated	omega_2	ZV-Lasso-5	0.1595	1.71e-04	0.0795	459.7714	17.7000
simulated	omega_2	ZV-OLS-1	0.1588	0.0012	0.0788	8.8343	3.0000
simulated	omega_2	ZV-OLS-2	0.1591	3.60e-04	0.0791	103.8631	9.0000
simulated	omega_2	ZV-OLS-5	0.1595	1.03e-05	0.0795	1.28e+05	55.0000
simulated	omega_3	MC	0.7306	0.0080	0.1496	1.0000	0.0000
simulated	omega_3	ZV-Lasso-5	0.7350	3.79e-04	0.1450	445.9123	19.3000
simulated	omega_3	ZV-OLS-1	0.7349	7.44e-04	0.1451	115.5784	3.0000
simulated	omega_3	ZV-OLS-2	0.7347	4.78e-04	0.1453	279.6720	9.0000
simulated	omega_3	ZV-OLS-5	0.7349	4.52e-05	0.1451	3.13e+04	55.0000

Table 1: Repeated-run summary for simulated data. "sd across runs" is the empirical numerical error across the 10 independent chains.

The first-order controls already reduce variance substantially. Degree-2 controls improve further. Degree-5 OLS is extremely effective on this low-dimensional simulated example, but this is also the setting where it is easiest for OLS to exploit the full dictionary without numerical problems. Post-Lasso keeps fewer controls and still gives a large variance reduction.

5.2 EUR/USD Data

data	param.	method	mean	sd across runs	rmse/ref.	var. red.	avg. selected
eurusd	omega_1	MC	0.0864	0.0187	0.0178	1.0000	0.0000
eurusd	omega_1	ZV-Lasso-5	0.0783	0.0159	0.0171	1.3776	36.8000
eurusd	omega_1	ZV-OLS-1	0.0853	0.0173	0.0164	1.1742	3.0000
eurusd	omega_1	ZV-OLS-2	0.0770	0.0062	0.0111	9.0143	9.0000
eurusd	omega_1	ZV-OLS-5	0.0773	0.0220	0.0227	0.7258	55.0000
eurusd	omega_2	MC	0.0856	0.0113	0.0107	1.0000	0.0000
eurusd	omega_2	ZV-Lasso-5	0.0910	0.0047	0.0070	5.6890	26.6000
eurusd	omega_2	ZV-OLS-1	0.0905	0.0051	0.0069	4.8051	3.0000
eurusd	omega_2	ZV-OLS-2	0.0894	0.0045	0.0057	6.3006	9.0000
eurusd	omega_2	ZV-OLS-5	0.0895	0.0058	0.0068	3.7623	55.0000
eurusd	omega_3	MC	0.4624	0.0937	0.0889	1.0000	0.0000
eurusd	omega_3	ZV-Lasso-5	0.5017	0.0992	0.1020	0.8923	38.1000
eurusd	omega_3	ZV-OLS-1	0.4639	0.0918	0.0871	1.0417	3.0000
eurusd	omega_3	ZV-OLS-2	0.5087	0.0319	0.0553	8.6527	9.0000
eurusd	omega_3	ZV-OLS-5	0.5072	0.1091	0.1128	0.7373	55.0000

Table 2: Repeated-run summary for EUR/USD returns. The reference for the RMSE column is the pooled raw-MC mean.

For the real-data example the full degree-5 OLS estimator is less reliable: for ω_1 and ω_3 its repeated-run variance is worse than MC. This matches the conditioning issue: a larger dictionary can help, but the naive OLS fit may become unstable. Degree-2 OLS is robust here, and the Lasso-screened estimator improves ω_2 while avoiding the full degree-5 regression.

6 MCMC Dependence and Regression

The linear regression used to estimate control-variate coefficients is not literally an iid regression because the inputs are MCMC draws. The coefficient estimate can still be consistent under ergodic-average conditions, but the usual iid regression standard errors are not valid. We addressed this in two ways:

- numerical error is reported through repeated independent chains, which directly assesses estimator variability;
- batch-means standard errors are computed in the notebook for the controlled output series.

Other practical solutions are to estimate coefficients on a pilot chain and evaluate on a second independent chain, to fit on thinned samples, or to use block cross-validation for Lasso.

parameter	full_sample_estimate	full_batch_se	thin5_estimate	thin5_batch_se
omega_1	0.107823	0.000042	0.107731	0.000027
omega_2	0.159526	0.000005	0.159530	0.000003
omega_3	0.735131	0.000041	0.735224	0.000025

Table 3: A small thinning check on the simulated representative chain. Thinning changes little here, suggesting that the regression fit is not dominated by adjacent highly dependent draws.

7 Conclusion

The project confirms the main message of zero-variance control variates. Once $\nabla \log \pi$ is available, first-order polynomial controls are easy to add to an MCMC estimator and can sharply reduce numerical error. Higher-order controls can reduce the variance further, but the OLS regression can become ill-conditioned. Lasso screening is a practical compromise: it lets us start from a larger control dictionary without always paying the numerical price of the full regression.

References

[1] A. Mira, R. Solgi, and D. Imparato. Zero variance Markov chain Monte Carlo for Bayesian estimators. *Statistics and Computing*, 23:653–662, 2013. doi:10.1007/s11222-012-9344-6.

[2] R. Leluc, F. Portier, and J. Segers. Control variate selection for Monte Carlo integration. *Statistics and Computing*, 31:50, 2021. doi:10.1007/s11222-021-10011-z.

[3] Frankfurter exchange-rate API. `https://www.frankfurter.app/docs/`.